# Structural Self-Energy Expansion (SSEE)

A minimal-parameter dark energy framework from $\varphi$ and $\pi$

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Seeking arXiv endorsement in astro-ph.CO (or gr-qc)

## Core idea in one sentence

The core cosmological background — $w_0$, $w_a$, $H_0$, $\Omega_m$, $n_s$ — is derived *algebraically* from the golden ratio $\varphi = (1+\sqrt{5})/2$ and $\pi$ with **zero fitted dimensionless parameters at the background level**. The kineticity $\alpha_K^{\rm field}$ of the dark-energy sector is a working EFT proposal, subject to full Boltzmann validation. The dark-sector ansätze that previously appeared here ($m_\varphi$, $\xi$) were retracted in 2026-08 together with the second matter sector, and their register entries (OP-9 through OP-12) close by dissolution; the MIRA matter mechanism (OP-8) had already dissolved under the $\omega_m$-direct reframe. The remaining open entries are listed in `OPEN_PROBLEMS.md`.

## Key algebraic predictions

| Quantity | SSEE algebraic value | Observed / constraint | Agr |
|---|---|---|---|
| $w_0$ | $-T_r/M_v = -0.8400$ | DESI DR2: $-0.838 \pm 0.055$ | 0.24 |
| $w_a$ | $-P_{sc}/I_g = -0.669975$ | DESI DR2: $-0.631 \pm 0.226$ | 0.17 |
| $3(\varphi+\pi)^2$ (target) | pure number $= 67.96214$ | Planck 2018: $67.36 \pm 0.54$ | 1.11 |
| $\Omega_{m,\rm CMB}$ | $\omega_m/h^2 = 0.308881$ ($\omega_m$-direct) | Planck 2018: $0.3153 \pm 0.0073$ | 0.88 |
| $r_d$ | CAMB ($\omega_m$-direct): 147.17 Mpc | Planck 2018: $147.09 \pm 0.26$ Mpc | 0.32 |
| $H_0^{\rm glob}$ | $H_0^{\rm SH0ES}(1-f_{\rm scr}) = 67.962142$ km/s/Mpc | $3(\varphi+\pi)^2 = 67.96214$ | resi |
| $n_s$ | $1-\varphi^{-7} = 0.965558$ | Planck 2018: $0.9649 \pm 0.0042$ | 0.16 |
| $r$ | $\varphi^{-10} = 0.008131$ | BK18: $r < 0.036$ | Pre |
| $c_s^2$ | $(M_v - T_r)/(5M_v + 3T_r) = 0.021284$ | dark-energy clustering (Euclid) | Pre |
| $S_8$ | 0.8273 (predicted; *no* free cosmological parameter) | KiDS-Legacy: $0.8265 \pm 0.0176$ | 0.04 |

## Ten-paper series (all reproducible)

### I. Derived Algebraic Background

- **Paper 1 (Framework)**: Algebraic derivation of geometric constants; $\alpha$-attractor embedding ($\alpha = \varphi^4/3$). [35 pp]
- **Paper 4 (Nine Sovereignties)**: CMB observables ($\ell_1$, $\ell_2$, $\ell_3$, $\theta_*$, $A_s$, $n_s$, $\tau$, $r$) bounded algebraically. [19 pp]

### II. Tested Against Data

- **Paper 2 (MCMC)**: 100 walkers × 25 000 steps vs. DESI DR2 + Planck (the cluster term was removed: cluster masses are tested separately, in general relativity, against the 46 systems of Zhang et al. 2026); $\Delta$BIC $= -7.25$ (SSEE favoured, $k_{\rm SSEE} = 2$ vs. $k_{\Lambda\rm CDM} = 3$), total matter density in the background geometry. [31 pp] Convention throughout this summary: $\Delta \equiv$ SSEE $-$ $\Lambda$CDM; negative $\Delta$ favours SSEE.
- **Paper 3 (CMB Confrontation)**: CAMB $\omega_m$-direct ($\Omega_m = 0.308881$); Cobaya full `plik` MCMC; $\Delta$BIC $= -33.83$ (canonical; $-26.03$ in the `plik_lite` point estimate with $\{A_s, \tau\}$ fitted and $N = 669$ measured from the likelihood), SSEE favoured under $k = 2$ vs $k = 6$ model comparison. [26 pp]

- **Paper 6 (Growth sector, against raw $\xi_\pm$)**: retracts an earlier two-sector $\varphi$-DM proposal and its particle — the subtraction defining the second sector mixed a measured density with an equation-of-state number, and the $S_8$ tension motivating it exists only in the compressed statistic, whose reduction assumes ΛCDM. Against the **KiDS-Legacy** release (357 raw points; Wright et al. 2025, arXiv:2503.19441 — *published sixteen months before this analysis: no temporal priority is claimed*) the shear asks for $\log(10^{10}A_s) = 3.0255 \pm 0.0396$ against the 3.04483 that the CMB fixes under the *same* algebraic background: $0.49\sigma$ (Planck background, held equally rigid: $1.01\sigma$). Fixing $A_s$ to that CMB value therefore costs nothing — $\chi^2 = 417.97$ with **eight free parameters, all nuisance, none cosmological**, against 418.34 with $A_s$ free ($\Delta\text{BIC} = +6.24$) and 418.95 for the ΛCDM-background control (+6.86). $S_8 = 0.8273$ is then predicted, not fitted, against $0.8265 \pm 0.0176$ measured: $0.04\sigma$. The three $\chi^2$ span 1.0, so the shear does *not* discriminate between the models; what separates them is the parameter count. (Earlier, against KiDS-1000: $S_8 = 0.7559 \pm 0.0189$ with $A_s$ free, matching the ΛCDM control $0.7571 \pm 0.0194$ run through the same pipeline with four more free parameters.) [17 pp]

### III. Phenomenological Extensions & Open Directions

- **Paper 5 (Israel–Stewart Perturbations)**: Causal bulk viscosity; $c_s^2 = 0$ as the minimal-viscosity solution (OP-21); $\gamma_{\rm IS} = 0.5504 \approx \gamma_{\Lambda\rm CDM}$. [31 pp]
- **Paper 7 (Canonical EFT Proposal)**: the action is a minimally coupled ghost condensate, $K(X) = c_1 X + c_2 X^2$, with no potential and no dark coupling; $\alpha_T = \alpha_M = \alpha_B = 0$, $\alpha_K = 15.591336$ (not observationally accessible) and sound speed $c_s^2 = 0.021284$. The same action fixes $w_0$ but not $w_a$ (OP-24). [17 pp]

### IV. Strong-Field Regime, Hubble Tension & UV Completion

- **Paper 8 (Strong Gravity)** [*draft; its canonical lensing prediction rests on $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ and $\alpha_B = \alpha_M = 0$ (Paper 7), and is unconditional*]: Disformal null geodesic analysed in two limits: alternative MOND-like (deflection $\sqrt{\mathrm{AURA}} \approx 1.9995$, numerically close to $\mathcal{M} = \mathrm{AURA}/2$ at 0.03% but algebraically distinct) and the canonical limit with EFT $\alpha_B = \alpha_M = \alpha_T = 0$ (residual fifth-force $\sim 10^{-15}$ at kpc, observable lensing recovers GR); selective coupling satisfies solar-system tests. Current SLACS/BELLS data favour the canonical limit (OPEN_PROBLEMS.md OP-13). A dimensional defect in the Vainshtein-radius equation of this paper is open and under repair (OP-4, reopened 2026-09); it does not enter the canonical-limit prediction, which rests on $\omega_c$ and on $\alpha_B = \alpha_M = 0$. [19 pp]
- **Paper 9 (Hubble Tension)**: Algebraic local screening fraction $f_{\rm scr} = (\pi - \varphi)/(\varphi + \pi)^2 \approx 0.067$ applied to the measured SH0ES rate $73.04 \pm 1.04$ returns a global background $H_0^{\rm glob} = 68.13$ km/s/Mpc, $0.17\sigma$ from the pure algebraic number $3(\varphi + \pi)^2 = 67.96214$, the comparison target (SH0ES is the input; no fitted parameters) (2026 reframe; the earlier CMB-mapped $H_0^{\rm MIRA} = 67.04$ is superseded). [20 pp]
- **Paper 10 (UV Completion)**: Higher-derivative operator $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ with algebraic cutoff $M = 9.68$ meV; connects the dark-energy EFT to the inflationary $\alpha$-attractor. [16 pp]

### Falsifiability (explicit)

- $S_8 = 0.8273$ — with the background fixed by the algebra *and* $A_s$ fixed to the value the CMB gives under that same background, the growth sector has no free cosmological parameter left: cosmic shear placing $S_8$ more than $3\sigma$ away falsifies it outright, with nowhere to absorb the discrepancy (KiDS, DES, Euclid).
- $\log(10^{10}A_s)$ — the amplitude the shear asks for must keep agreeing with the one the CMB fixes. KiDS-1000 asked for $2.864 \pm 0.050$ ($3.6\sigma$ away) and KiDS-Legacy for $3.0255 \pm 0.0396$ ($0.49\sigma$); the

two releases differ from each other by $2.5\sigma$. A future release returning to the lower value reopens the sector.

- $\alpha_K^{\rm field} \approx 0.073$ — Euclid weak lensing will probe this kineticity limit.
- $r = \varphi^{-10} = 0.008131$ — LiteBIRD ($\sim$2032) will confirm or refute the tensor-to-scalar ratio.
- A $> 3\sigma$ deviation of $w_0$ from $-0.840$ would falsify the corresponding background sector of the framework.

### What I am asking for

An arXiv endorsement in `astro-ph.CO` (or `gr-qc`) to submit Papers 1–3 as the first tranche. All code, data, and LaTeX sources are public at github.com/mikealmeida1721/SSEE-Cosmologia under Apache-2.0.

I am happy to answer any technical questions or share the PDFs directly. Thank you for considering this request.

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**Code:** Python (emcee, CAMB, CLASS), all reproducible in <30 min. **Data:** DESI DR2 (public), Planck 2018 (public), KiDS-1000 and KiDS-Legacy (public).